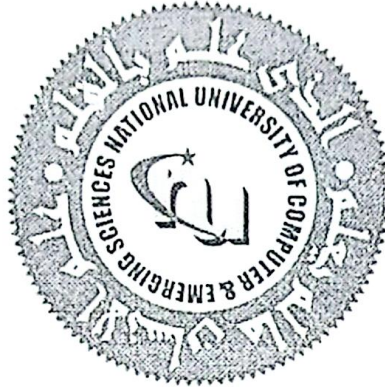


FAST- National University of Computer and
Emerging Sciences



CNET

Assignment # 2

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SECTION: BS(CS) - F

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Q2:

Data:

$$S = 2500 \text{ km} = 2.5 \times 10^6 \text{ m}, \text{ prop-speed} = 2 \times 10^8 \text{ m/s}$$

$$R = 10 \text{ Gbps} = 10 \times 10^9 \text{ bps}$$

$$MSS = 12000 \text{ bits} \Rightarrow L = 12000$$

1. d-prop?, RTT=?

2. L/R =?

3. U =?

4. Factor increase in U' ?, 64-bit packet pipeline

Sol:

$$\begin{aligned} \therefore \text{d-prop} &= \frac{S}{\text{prop-speed}} \\ &= \frac{2.5 \times 10^6}{2 \times 10^8} \\ &= 0.0125 \text{ s} \end{aligned}$$

$$\boxed{\text{d-prop} = 12.5 \text{ ms}} \text{ Ans.}$$

$$\begin{aligned} \therefore \text{RTT} &= 2 * \text{d-prop} \\ &= 12.5 * 2 \end{aligned}$$

$$\boxed{\text{RTT} = 25 \text{ ms}} \text{ Ans.}$$

$$\begin{aligned} \Rightarrow L/R &= \frac{12000}{10 \times 10^9} \\ &= 1.2 \times 10^{-6} \text{ s} \end{aligned}$$

$$\boxed{L/R = 1.2 \mu\text{s}} \text{ Ans.}$$

$$\therefore U = \frac{L/R}{\text{RTT} + (L/R)}$$

$$\Rightarrow U = \frac{1.2 \times 10^{-6}}{25 \times 10^{-3} + 1.2 \times 10^{-6}}$$

$$\Rightarrow \boxed{U = 4.8 \times 10^{-5}} \text{ Ans.}$$

$$\Rightarrow U = 0.0048 \%$$

So, if we have $N = 64$

$$\therefore U' = \min(U * N, 1)$$

$$\Rightarrow U' = \min(4.8 \times 10^{-5} * 64, 1)$$

$$U' = \min(3.07 \times 10^{-3}, 1)$$

$$\boxed{U' = 3.07 \times 10^{-3}} \text{ ~~Ans.~~$$

$$U' = 0.3 \%$$

$$\text{Factor increase} = U'/U$$

$$= 3.07 \times 10^{-3} / 4.8 \times 10^{-5}$$

$$\boxed{\text{Factor inc.} = 64} \text{ Ans.}$$

②

Q3:

Data:

Pay load size = 1000 bytes

ISN-A = 10000

Order recieved = 2nd, 3rd, 1st

Segment 1:

Seq. No = 10000

Bytes contained = 10000 - 10999

Segment 2:

Seq. No = 11000

Bytes = 11000 - 11999

Segment 3:

Seq No = 12000

Bytes = 12000 - 12999

Sol:

1- Segment 2 arrives:

Buffer = {[11000 - 11999]}

Cumulative ACK sent: 10000 (still expecting 1st segment)

Highest contiguous byte = 9999

2- Segment 3 arrives:

Buffer = {[11000 - 11999], [12000 - 12999]}

Cumulative ACK sent: 10000

Highest contiguous byte = 9999

3- Segment 1 arrives:

Buffer = { }

Cumulative ACK sent: 13000

Highest contiguous byte = 12999 (All of the contiguous data was present so rest of the buffer was also proceeded).

Q4:

Data:

One way delay = d

Window size = 4

Timeout (T) = $4d$

To deliver = $P_0, P_1, P_2, P_3, P_4, P_5$

Data lost = P_1 & P_3

ACK delay = ACK 0 & ACK 2 by $3d$

Transmissions $\rightarrow$ $T_r = 0$ (Initially) ^{Others by d}

Table of actions (GBN):

Time | Actions

0	P_0 to P_3 transmitted ($T_r = 4$)
$1d$	P_0 & P_2 reach destination, ACK(0) sent, P_1 & P_3 lost
$2d$	ACK(0) on way
$3d$	"
(T) $4d$	Timeout, ACK(0) reach source, (but time out had priority so) P_0 to P_3 transmit, window slides one block, P_4 is transmitted ($\because$ Serialization is negligible so we'll consider P_4 to be transmitted with $P_0 - P_3$) ($T_r = 5$)
$5d$	P_0 to P_4 reach source, ACK(0-4) emitted
$6d$	ACK 1, 3, 4 reach source, window shifts, P_5 transmitted ($T_r = 1$)
$7d$	ACK 0, ACK 2 on way, P_5 reach source, ACK 5 emitted
(T) $8d$	Timeout, ACK 0, ACK 2 reach source, ACK 5 reach source, P_5 transmits again. ($T_r = 1$)

Total Data Transmissions = $4 + 5 + 1 + 1$

" = 11

Time when final delivery completes (reaches destination) = $7d$

4

Actions Table (SR):

Time	
0	P0 to P3 transmitted ($T_r + 24$)
1d	P1 & P2 reach destination; ACK(0) sent, P1 & P3 lost
2d	ACK(0) on way
3d	"
(T) 4d	Timeout for all, ACK(0) reach source, P0 to P3 transmit, window slides one block, P4 transmitted ($T_r + 5$)
5d	P0 to P4 reach source, ACK(0-4) emit
6d	ACK 1, 3, 4 reach source, window shifts, P5 transmitted ($T_r + 1$)
7d	ACK 0, 2 on way, P5 reach source, ACK 5 transmit
(T) 8d	Timeout(P0-P4), ACK(0 & 2) reach source, ACK 5 reach source

Total data Transmission = $4 + 3 + 1 = 10$: Ans

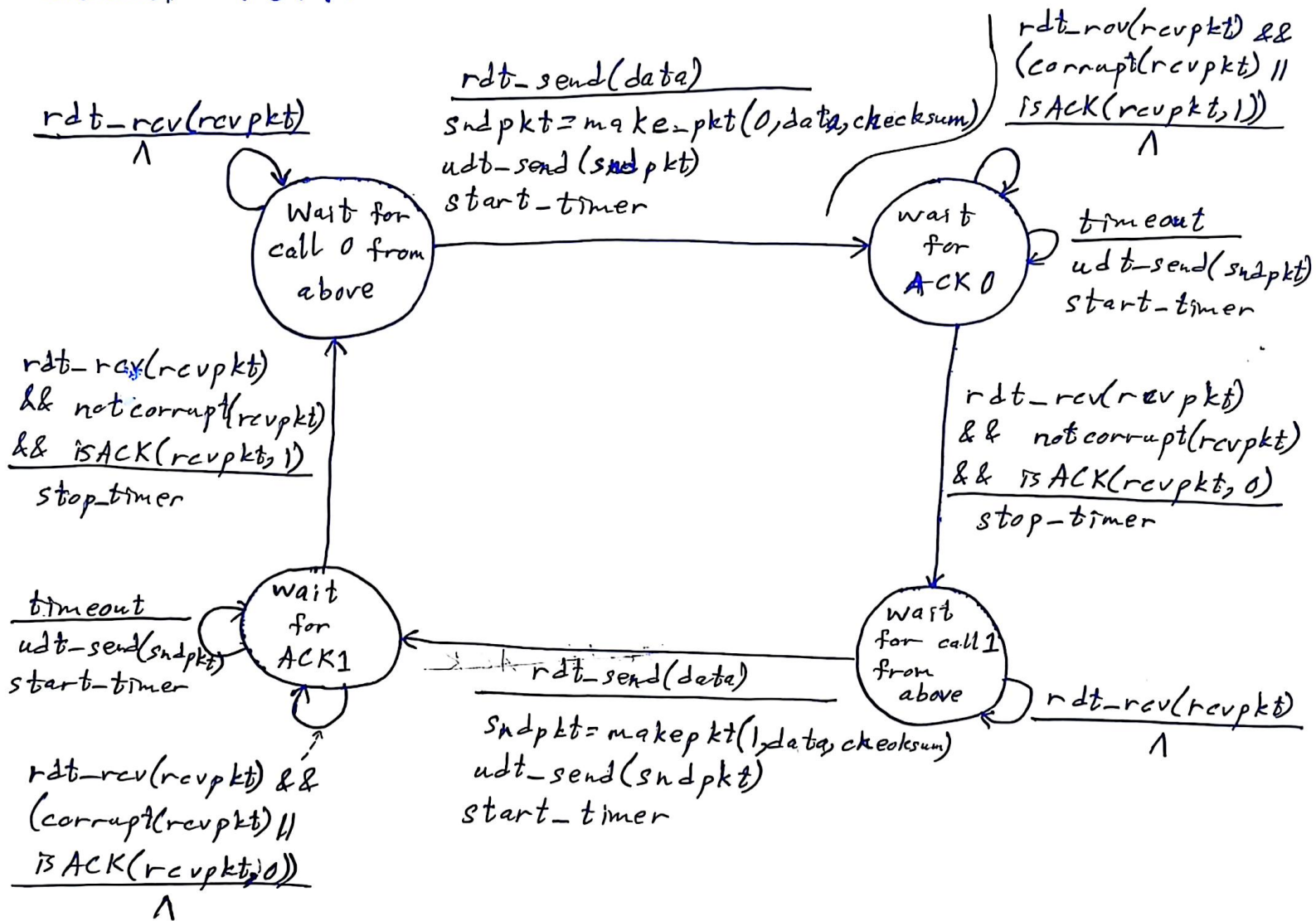
Time = 7d : Ans

Q5:

Step	Packet Arrives	Receiver State (Buffer)	Sent ACKs	Delivered
1	Packet 0	[]	ACK 0	0
2	" 2	[2]	ACK 2	-
3	" 4	[2, 4]	ACK 4	-
4	" 1	[2, 4]	ACK 1	1, 2
5	" 3	[4]	ACK 3	3, 4

Q1:

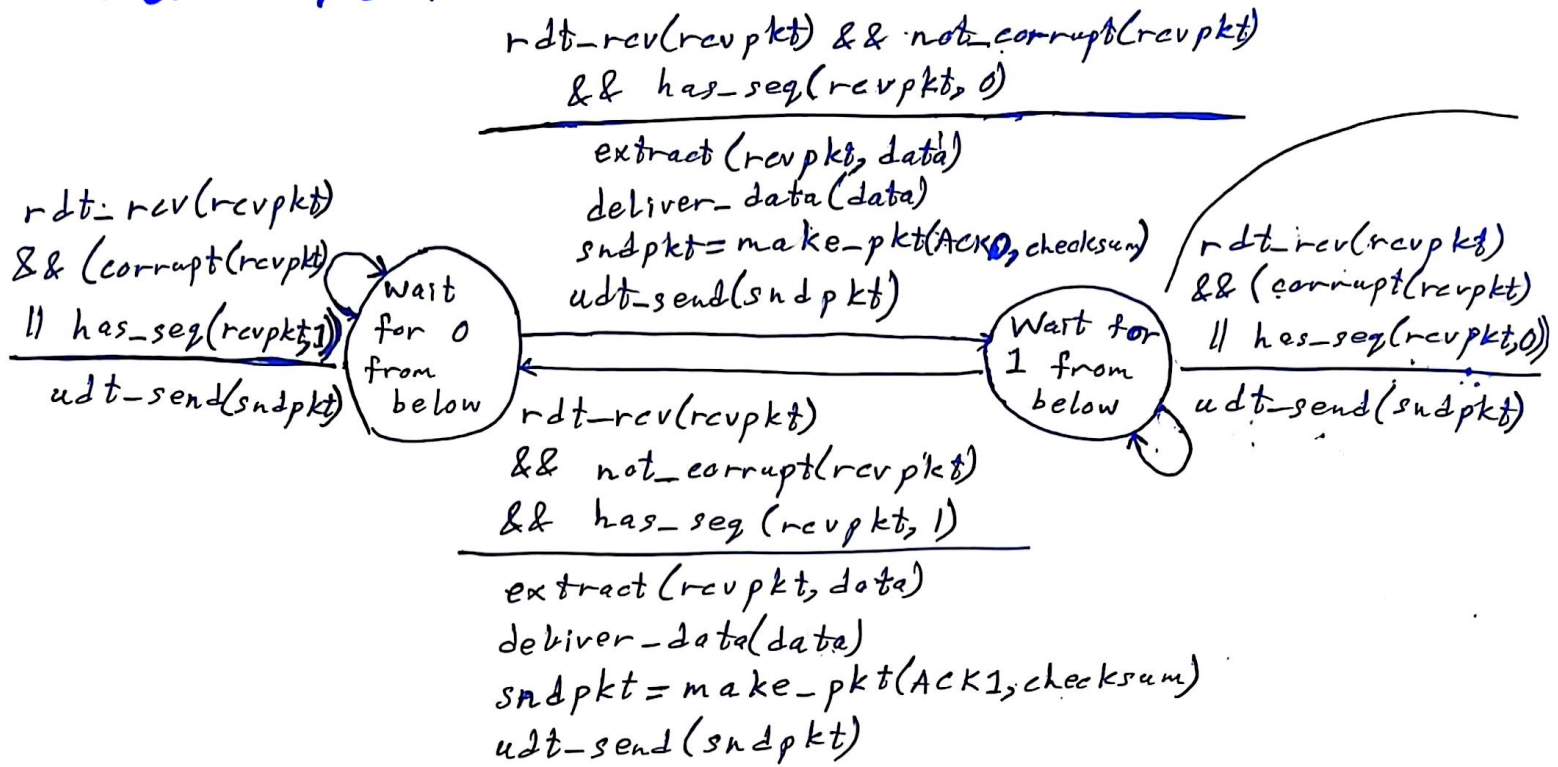
1.
Sender FSM:



6

2.

Receiver FSM:



3.

- receives packet 0
- "Wait for 0" state receiver sends ACK[0]
- "Wait for 1" state receiver sends ACK[1]
receives packet 1
- "Wait for 1" state receiver receives corrupt/duplicate packet 0
sends ACK[0]

Reason:

- Moves on to next, if receives packet correctly
- Duplicate ACKs signal that it is still waiting for next one
- NAK free it is. It just sends last correctly received ACK.

4.

Sender: Ignores out of order ACKs, If doesn't receive ACK within certain time, timeout happens and sender retransmits.

Receiver: Checks sequence number & discards duplicates, then re-ACKs last correct packet.

(Duplicate ACKs signal missing/corrupted packets.)

5.

As stated above, receiver makes sure no packet is delivered more than once by discarding duplicate & sending ACK.

